

William Harris

Robotics Engineer | Medical Device Design, Mechatronics, Haptic Systems

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EDUCATION

UC San Diego

MS: Mechanical Engineering | GPA: 3.68

BS: Mechanical Engineering | Minor: Design

La Jolla, CA

Expected **June 2027**

June 2025

SKILLS

Design & Fabrication: SolidWorks, CREO, Ansys FEA, Fusion 360 CAM, DFMA, 3D Modeling, Rapid Prototyping, Laser/Waterjet Cutting, CNC Machining; **Electronics & Controls:** Schematic Design (EasyEDA), Motor Control Systems, CAN Communication, BLDC Motors; **Software:** Python, MATLAB, Arduino C, ROS2, Embedded Linux, MuJoCo; **Specialized:** Soft Robotics & Compliant Mechanisms, Surgical Instrument Design

EXPERIENCE

Cor Medical Ventures — *Product Development Intern* — Del Mar, CA

Jun 2026 – Sep 2026

- Designed ICU calibration test fixtures, touch-alignment detection circuit, LED PCB, and quick-release endoscope latch for tendon-driven endoscopic surgical robot
- Redesigned base station console enclosure, wiring, and button layout to improve surgeon usability
- Developed Python duty-cycle analysis script from robot motor logs and live OR execution data
- Redesigned 5 surgical instruments from metal to single-use injection-molded toolkit using DFMA; reduced one assembly from 5 to 3 parts (40%); designed Medtronic Navloc and standard variants
- Reviewed DHR documentation, identifying out-of-spec shipped items to ensure FDA compliance

Advanced Robotics and Controls Lab (ARCLab) — *Graduate Researcher* — La Jolla, CA

Jun 2024 – Present

- Adaptive Morphing Leg for Search and Rescue Robots: Designed and built 5-bar linkage leg testbed with reconfigurable geometry for search and rescue operations; optimized link lengths to increase foot force by 83% (58–106 N); developed non-backdrivable capstan-drive actuator with Python/CAN control for gait coordination.
 - Completing single-leg force optimization analysis; planning a bilevel RL framework to co-optimize geometry and locomotion policy for rescue and search modes on the full hexapod.
- Mobile Manipulator Platform for Rescue Operations: Built structural frame and linear capstan actuator for mobile manipulator supporting two 7-DoF Franka Panda arms (36 kg, 10 kg payload); designed BLDC-driven translation stage to shift center of mass, eliminating 60 kg counterweight (30% weight reduction) and enabling human-safe physical interaction testing.
 - Designed fin-ray-effect TPU soft gripper for compliant human limb grasping (inspired by Stanford UMI)
- PHABS (Portable Haptic Assisted Bimanual System): Developed lateral force-feedback mechanism for handheld bimanual haptic teleop device with spring-damper contact model; pilot testing showed haptic feedback improved manipulation task success 3× and enabled real-time grip awareness.

Triton Robotics — *President ('23-'24); Lead Mechanical Engineer ('21-'23)* — La Jolla, CA

Sep 2021 – Jun 2024

- Led redesign of projectile-launching robot, improving rotation speed by 40% and targeting accuracy by 50% through new drivetrain and slip-ring turret; mentored six members and presided over 80-person team.

PROJECTS

Particle-Jamming Limb Stabilization for Rescue Robotics — *Individual Project*

Dec 2025

- Designed and tested a vacuum-tunable particle-jamming interface for emergency medical limb stabilization
- Built and calibrated a custom FSR array to assess contact engagement; pull-out testing showed the jamming pad achieved 1.5–4× greater extraction resistance than strap-based splints depending on loading conditions.

PUBLICATIONS

- Co-first Author**, ICRA 2026 (Accepted): "[Dynamically Extensible and Retractable Robotic Leg Linkages for Multi-task Execution in Search and Rescue Scenarios](#)." arXiv:2511.10816